



Cambridge O Level

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COMPUTER SCIENCE

2210/13

Paper 1 Computer Systems

October/November 2024

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **12** pages. Any blank pages are indicated.



1 Data can be measured using different units of storage.

(a) Tick (✓) **one** box to show which of the following is the largest unit of data storage.

A tebibyte (TiB) ☐

B pebibyte (PiB) ☒

C mebibyte (MiB) ☐

D gibibyte (GiB) ☐

B
KiB
MiB
GiB
TiB
PiB

[1]

(b) A computer has primary storage.

Give **one** example of primary storage.

Explain the purpose of your chosen example.

Example RAM

Explanation Stores currently running programs and data; volatile in nature.

[3]

(c) All data is converted to binary to be processed by a computer.

(i) Calculate the binary number for the denary number 175. Show all your working.

$$(175)_{10} = (10101111)_2$$

128	64	32	16	8	4	2	1
1	0	1	0	1	1	1	1

[2]

$$\begin{array}{r}
 1 \\
 128 \\
 32 \\
 \hline
 160 \\
 15 \\
 \hline
 175
 \end{array}$$



(ii) Give the binary number for the given hexadecimal numbers.

15

0001 0101

2D

0010 1101

091

0000 0000 1001 0001

Working space

8 4 2 1
1 1 0 1

A 10

B 11

C 12

D 13

E 14

F 15

[3]

(d) Binary integers can be added together.

Add the **two** binary integers using binary addition. Show all your working. Give your answer in binary.

$$\begin{array}{r} 11100011 \\ + 11001100 \\ \hline 11010111 \end{array}$$

$$\begin{array}{r} 1 \\ 110 \\ 101 \\ 011 \end{array}$$

[4]

(e) Calculate the denary number for the two's complement binary integer 10001110. Show all your working.

$$\begin{array}{r} 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 10001110 \Rightarrow -114 \\ 1 -ve \quad 10001110 \\ 0 +ve \quad 01110010 \end{array}$$

$$\begin{array}{l} -128 + (8 + 4 + 2) \\ -128 + 14 = -114 \end{array}$$

[2]





- 2 An employee has a report that they need to email to their employer.

The employee compresses the report file before emailing it.

- (a) State the effect the compression has on the report file.

The file size is reduced. [1]

- (b) Give **two** benefits of compressing the report file before emailing it.

- 1 Faster sending: smaller file is equal to faster upload and download.
Less storage space used: save the space on email server and devices.
Fits into email size limits: some email services limit at attachment sizes.
Uses less Internet data: especially important on mobile data or slow connections.
 - 2 Less bandwidth usage: useful when many users are sharing the same network.
- [2]

- (c) The employee decides to use lossless compression to compress the file.

Explain why lossy compression is **not** suitable.

Lossy compression would permanently remove data from the report.
This can cause important text and formatting to be lost or damaged.
Reports need to maintain full accuracy, so lossy compression is not appropriate

[3]

- (d) When the employee enters the email address, the computer uses Unicode to convert the email address to binary.

- (i) State what Unicode is an example of.

Character set [1]

- (ii) Give **two** advantages of the computer using Unicode instead of American standard code for information interchange (ASCII).

- 1 More characters: UNICODE can handle way more characters than ASCII.
Supports multiple languages: Unicode includes Arabic, Chinese, Urdu, emojis and many world languages.
Include symbols and emojis: modern communication needs symbols, emojis, mathematical symbols, et cetera.
 - 2
- [2]



- (iii) Give **one** disadvantage of the computer using Unicode instead of ASCII.

So each character in uni code requires more storage space than ASCII. So it affects size of the file.

[1]

- (e) The report is broken down into packets to be emailed.

- (i) Circle **three** items of data that can be found in the packet header.

trailer

originator's address

payload

interrupt

input

operating system

destination address

antivirus

packet number

[3]

- (ii) Each packet is sent along a different path from the employee's device to the employer's device.

Tick (✓) **one** box to show the name of this method of sending packets.

A packet networking

☐

B packet circuiting

☐

C packet switching

☒

D packet transferring

☐

[1]

- (iii) A hardware device is used to control the path that each packet takes.

Give the name of this hardware device.

- Router -

[1]

- (f) The email data is checked for errors after it has been transmitted, using an echo check and a checksum.

- (i) Explain how the echo check is used to check for errors in the email data.

A copy of the sent data is returned back to the sender.

The sender compares the original data with the return copy.

If they do not match, an error has occurred during transmission.

[3]





- (ii) In the checksum error detection method, two values are compared after transmission. If the values do **not** match, an error is detected.

Explain why the values **not** matching would show an error has occurred.

The checksum is calculated again from the receive data using the same method.

If the calculated check sum is different from the original, it shows that the data was altered during the transmission.

.....

 [2]

- (g) The email data is encrypted using asymmetric encryption before it is sent.

- (i) Give **one** reason why the email data is encrypted.

To keep the email data secure and prevent unauthorised access.

.....
 [1]

- (ii) Give **one** similarity between symmetric encryption and asymmetric encryption.

Both methods use keys to scramble data means encrypting data into an unreadable form.

..... [1]

- (iii) Give **two** differences between symmetric encryption and asymmetric encryption.

1 First difference: symmetric encryption uses one key, whereas asymmetric encryption uses two keys that is a public key and a private key.

.....

2 Second difference: in symmetric encryption, the same key is used for both encryption and decryption, but in asymmetric encryption, different keys are used for encryption and decryption.

..... [2]

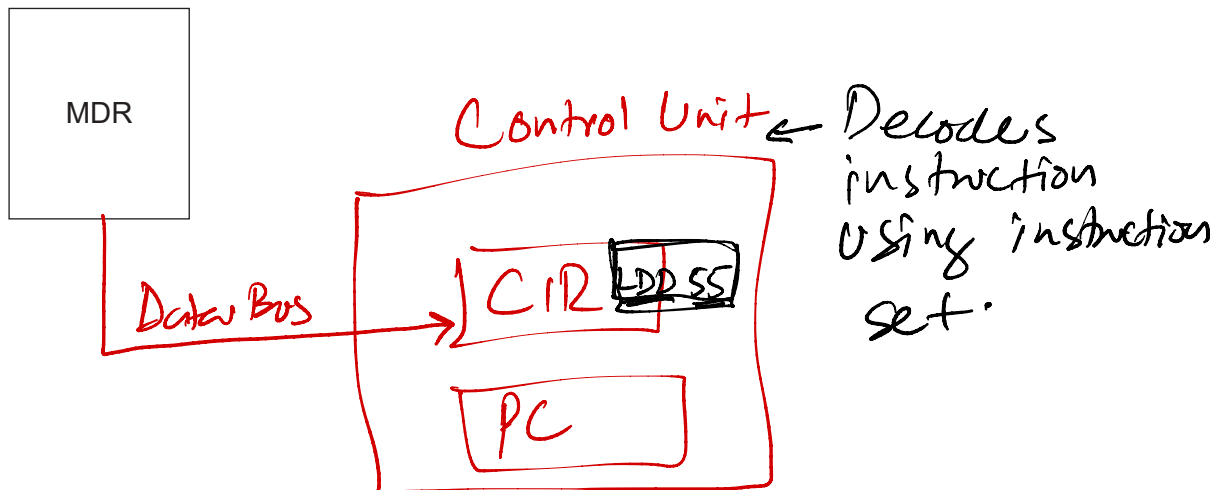


- 3 An instruction is fetched from random access memory (RAM) into the memory data register (MDR) to be decoded.

(a) Identify **two** other registers that are used in the fetch stage of the cycle.

- 1 Program Counter (PC)
 - 2 Mem. Add. Register (MAR)
- [2] Acc

- (b) Complete and annotate the diagram to show how the data is decoded once it has been fetched into the MDR.



[4]



4 Complete the table to give the missing term or description for the internet terms.

Internet term	Description
World wide web (www)	the collection of web pages accessed using the internet
IP Add.	the address given to a device when it connects to the internet
web browser	App. that allows user to access web pages.
DNS	the hardware that stores a database of matching website and IP addresses
Firewall Proxy Server	a type of hardware that can be used to prevent a distributed denial of service (DDoS) attack
hacking	Accessing or gaining access of a computer system without owner's consent.

[6]



- 5 A farmer has a plough that is an automated system. The plough is used to dig the ground in a field to prepare it for planting seeds.

The plough uses sensors and a microprocessor to maintain a straight line when digging the ground.

- (a) State what is meant by an automated system.

An automated system performs actions automatically without human intervention.

..... [1]

- (b) Describe the role of the microprocessor in this process.

The micro processor receives data from the plough's sensors.

It analyses the data to check if the plough is maintaining a straight line.

If adjustments are needed, it sends control signal to correct the plough's direction automatically.

.....

 [3]

- (c) Give **two** benefits to the farmer of using an automated system for this purpose.

1 There are many benefits:

1. it frees the farmer to do other jobs while the plough operates automatically.

2. it can work continuously without needing brakes, increasing efficiency.

.....
 [2]

- (d) The plough uses artificial intelligence (AI) to navigate its way around the field.

Explain how the plough makes use of AI for this purpose.

The machine collects data from sensors about the field shape, landscape, and other obstacles.

It uses AI to analyse this data and learn the most efficient parts to plough.

Overtime, it adapts its behaviour to avoid mistakes and optimise its routes.

It develops a map of the field to navigate accurately without human input.

.....

 [4]





6 (a) Complete the statements about cookies.

Use only terms from the list.

Not all terms need to be used. Some terms may be used more than once.

binary close denary [✓] expire

hexadecimal image malware operating system

[✓] permanent [✓] persistent [✓] session sound

[✓] temporary [✓] web browser

Cookies are small text files that are stored by a web browser.

Session cookies are temporary text files that are deleted when the session/web browser is closed.

Persistent cookies are Permanent text files that are stored on a user's secondary storage device until they are manually deleted or they expire.

[7]

(b) Give **three** examples of the use of cookies.

- | | |
|--------------------------------|-----------------------------|
| 1 <u>Payment Details.</u> | 5 <u>Login Details</u> |
| 2 <u>User Preferences.</u> | 6. <u>Targeted adverts.</u> |
| 3 <u>User's personal data.</u> | |
| 4 <u>Shopping cart.</u> | |

[3]





- 7 A computer programmer uses an integrated development environment (IDE) when creating a computer program.

Explain the purpose of the IDE.

- An IDE allows a programmer to write, edit, and develop code in one place.
- It provides tools to test and debug the code to find and fix errors easily.
- It can translate the code into machine language for the computer to understand....
- It offers features like auto completion and syntax highlighting to make coding faster and reduce mistakes.

.....

.....

.....

.....

..... [4]





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